

**VIETNAM NATIONAL UNIVERSITY – HO CHI MINH CITY
INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING**



**WEB APPLICATION DEVELOPMENT
IT093IU
LAB 5 (Practice)**

Member List:

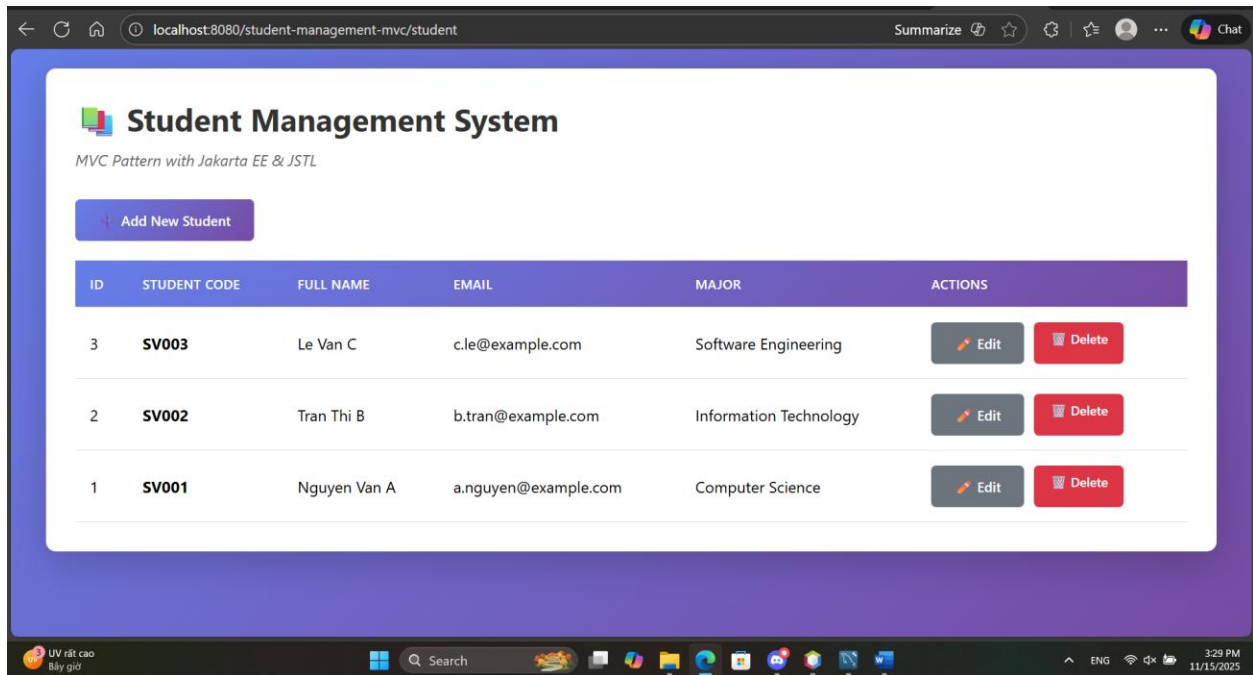
- Huỳnh Ngọc Bảo Long – ITITWE23023

Instructor: Ph.D. Lê Thị Ngọc Hạnh

Table of Contents

- List3
- Add3
- Edit6
- Delete8
- Empty State10

List



Scenario: User clicks "View All Students"

- Browser → GET /student?action=list
- Controller (StudentController) receives request
 - Extracts action=list parameter
 - Calls StudentDAO.getAllStudents()
- Model (StudentDAO) executes SQL
 - Returns List<Student>
- Controller sets data as request attribute
 - Forwards to student-list.jsp
- View (JSP) displays using JSTL
 - `<c:forEach items="${students}">`
- Browser receives HTML response

Add

Scenario: User click "Add new student" → Fill form → Submit

1. **Navigate to Add Form**
 - Browser → GET /student?action=new

- Controller (StudentController) receives request
 - Extracts action=new
 - Calls showNewForm()
 - Creates an empty form (no Student object)
- Controller forwards
 - /views/student-form.jsp
- View (JSP) renders blank form
 - No pre-filled values
 - Uses `${student == null}` paths
- Browser receives HTML form

The screenshot shows a web browser window with the address bar displaying `localhost:8080/student-management-mvc/student?action=new`. The page features a purple gradient background. In the center, there is a white card titled '+ Add New Student'. The card contains four form fields, each with a red asterisk indicating it is required:

- Student Code ***: A text input field with placeholder text 'e.g., SV001, IT123' and a note below it stating 'Format: 2 letters + 3+ digits'.
- Full Name ***: A text input field with placeholder text 'Enter full name'.
- Email ***: A text input field with placeholder text 'student@example.com'.
- Major ***: A dropdown menu with the text '-- Select Major --' and a downward arrow.

At the bottom of the card, there are two buttons: a purple button with a plus icon and the text '+ Add Student', and a grey button with a red 'X' icon and the text 'Cancel'.

2. Submit Add form

- Browser → POST /student with parameters
 - action=insert
 - studentCode=SV001
 - fullName=Test User
 - email=test@example.com
 - major=Computer Science
- Controller receives POST
 - Extracts action=insert

- Calls insertStudent()
- Creates new Student object with submitted form data
- Calls StudentDAO.addStudent(student)
- Model (StudentDAO) executes SQL
 - PreparedStatement INSERT
 - Adds row to students table
- Controller redirects
 - → student?action=list&message=Student+added+successfully
- Browser performs redirect → GET list
 - Controller loads updated list
 - Forwards to student-list.jsp
- View renders table
 - New student now appears
 - Success banner shown using `${param.message}`

localhost:8080/student-management-mvc/student?action=new

+ Add New Student

Student Code *

SV004

Format: 2 letters + 3+ digits

Full Name *

Jane Doe

Email *

janedoe@testmail.com

Major *

Business Administration

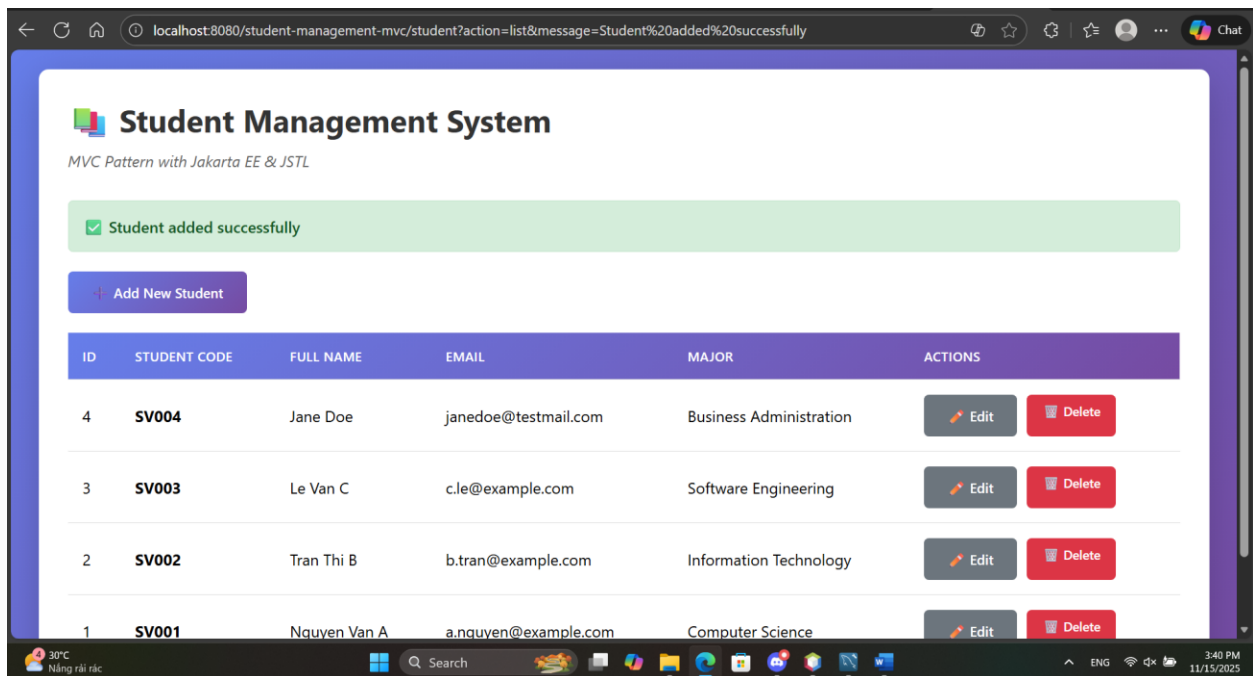
+ Add Student

✗ Cancel

30°C
Nắng rải rác

Search

ENG 3:40 PM
11/15/2025



Edit

Scenario: User click “Edit” → Modify form → Submit

1. Open Edit form

- Browser → GET /student?action=edit&id=5
- Controller receives request
 - Extracts action=edit and id=5
 - Calls StudentDAO.getStudentById(5)
- Model executes SQL
 - SELECT * FROM students WHERE id=?
 - Returns Student object
- Controller sets request attribute
 - request.setAttribute("student", studentObject)
 - Forwards to /views/student-form.jsp
- View (JSP) displays pre-filled form
 - Fields show the selected student info
 - `<input value="${student.fullName}">`

The screenshot shows a web browser window with the address bar displaying `localhost:8080/student-management-mvc/student?action=edit&id=4`. The main content area features a purple gradient background with a white central card titled "Edit Student" with a pencil icon. The form contains the following fields:

- Student Code ***: A text input field containing "SV004". Below it, a hint text reads "Format: 2 letters + 3+ digits".
- Full Name ***: A text input field containing "Jane Doe".
- Email ***: A text input field containing "janedoe@testmail.com".
- Major ***: A dropdown menu currently showing "Business Administration".

At the bottom of the card are two buttons: a purple "Update Student" button with a document icon and a grey "Cancel" button with a red 'X' icon. The Windows taskbar at the bottom shows the system clock as 3:52 PM on 11/15/2025.

2. Submit Edit form

- Browser → POST /student with parameters
 - action=update
 - id=5
 - Updated field values
- Controller receives POST
 - Extracts action=update
 - Creates updated Student object
 - Calls `StudentDAO.updateStudent(student)`
- Model (StudentDAO) executes SQL
 - `UPDATE students SET ... WHERE id=?`
- Controller redirects
 - → `/student?action=list&message=Student+updated+successfully`
- Browser follows redirect
- View shows updated list with success message

Edit Student

Student Code *
SV004
Format: 2 letters + 3+ digits

Full Name *
Jane Melisa Doe

Email *
janedoe@testmail.com

Major *
Information Technology

Update Student **Cancel**

Student Management System
MVC Pattern with Jakarta EE & JSTL

✓ Student updated successfully

Add New Student

ID	STUDENT CODE	FULL NAME	EMAIL	MAJOR	ACTIONS
4	SV004	Jane Melisa Doe	janedoe@testmail.com	Information Technology	Edit Delete
3	SV003	Le Van C	c.le@example.com	Software Engineering	Edit Delete
2	SV002	Tran Thi B	b.tran@example.com	Information Technology	Edit Delete
1	SV001	Nguyen Van A	a.nguyen@example.com	Computer Science	Edit Delete

Delete

Scenario: User click “Delete” → Confirm → Delete

- Browser → GET /student?action=delete&id=5 (after JavaScript confirm dialog)
- Controller receives request
 - Extracts action=delete, id=5

- Calls StudentDAO.deleteStudent(5)
- Model executes SQL
 - DELETE FROM students WHERE id=5
- Controller redirects
 - /student?action=list&message=Student+deleted+successfully
- Browser GETs the list again
- View renders updated table
 - Deleted student removed
 - Success banner shown

localhost:8080 says
Are you sure you want to delete this student?

OK Cancel

✓ Student updated successfully

Add New Student

ID	STUDENT CODE	FULL NAME	EMAIL	MAJOR	ACTIONS
4	SV004	Jane Melisa Doe	janedoe@testmail.com	Information Technology	Edit Delete
3	SV003	Le Van C	c.le@example.com	Software Engineering	Edit Delete
2	SV002	Tran Thi B	b.tran@example.com	Information Technology	Edit Delete
			a.nguyen@example.com	Computer Science	Edit Delete

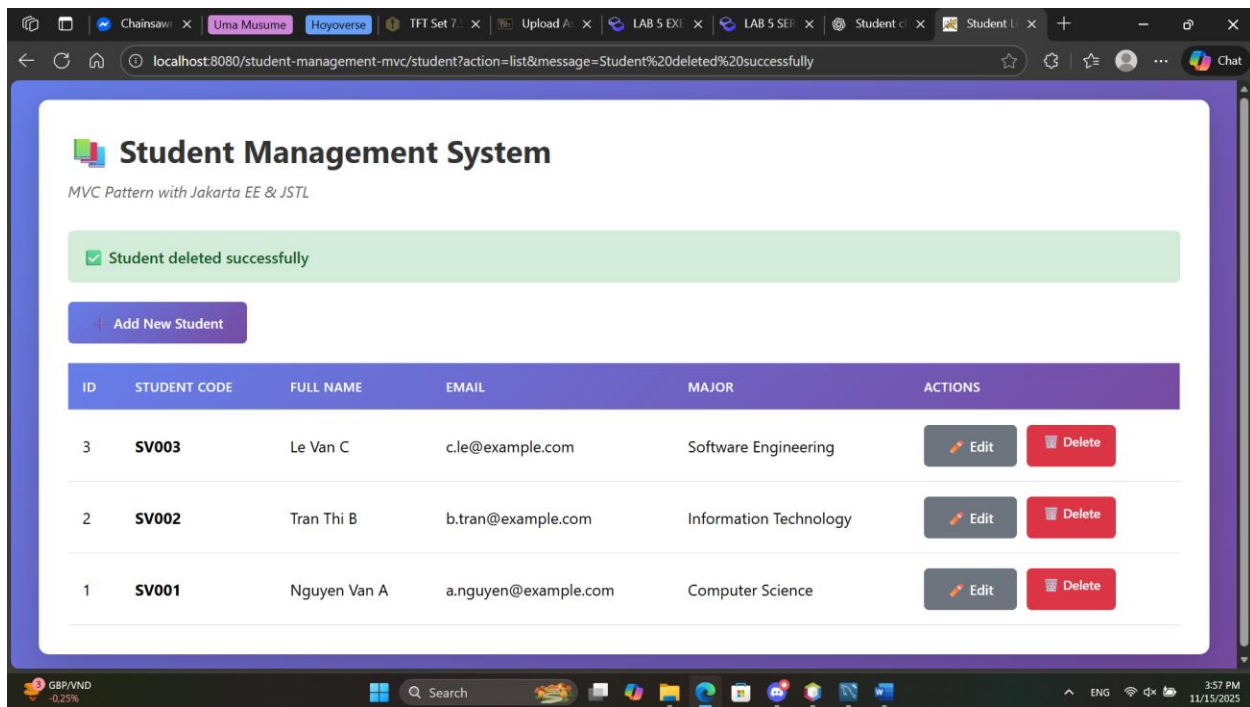
localhost:8080/student-management-mvc/student?action=list&message=Student%20updated%20successfully

localhost:8080/student-management-mvc/student?action=delete&id=4

GBP/VND -0.25%

Search

ENG 3:57 PM 11/15/2025



Empty State

Scenario: User deletes all students → View list

- Browser → GET /student?action=list
- Controller receives request
 - Calls StudentDAO.getAllStudents()
- Model executes SQL
 - SELECT * FROM students
 - Returns empty List<Student>
- Controller forwards
 - → /views/student-list.jsp
 - Sets request attribute: students = []
- View checks list
 - <c:when test="\${not empty students}">
 - → FALSE
- JSP displays empty state UI
 - “🚫 No students found”
 - A prompt to add new students
- Browser shows empty-state message

